

Lyapunov exponents spectrum estimation of fractional order nonlinear systems using Cloned Dynamics

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ABSTRACT

This work presents the determination of Lyapunov spectrum of fractional order dynamical systems. The method considers the Cloned Dynamic technique, which is based on time evaluation of exact copies of dynamic equations with small initial conditions disturbances. As its calculations are simple, this method can be applied to various dynamic systems types, in particular fractional order, nonlinear and time varying systems, and non linear systems where Jacobian are impossible to evaluate. The predictor-corrector Adams-Bashforth-Moulton algorithm is used for fractional order system numerical evaluation. All Lyapunov exponents are estimated based on the convergence or divergence of the small disturbed clones with respect to the reference (fiducial) trajectory and application of Gram-Schmidt Reorthonormalization after each convenient periodic time interval. To evaluate the application of this method, three third order dynamical systems are explored: Jerk, Financial, and a Four Wing systems. Integer order, commensurate fractional order and incommensurate fractional order are explored. To confirm chaotic or non-chaotic behavior, 0-1 Test is used and Jerk system is studied with a classical method for Lyapunov exponents determination also. The advantages and contribution of the proposed method the possibility to study incommensurate orders fractional cases and explore non linear systems without need of Jacobian determination, as is proved with applications shown. The Lyapunov Exponents and the phase portraits were simulated together to validate this method and the obtained results are in good agreement.

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1. Introduction

Fractional calculus which includes fractional integration and fractional differentiation is as old as its familiar counterpart, classical calculus or integer order calculus. It can be dated back to the Leibniz's letter to L'Hopital, dated 30 September 1695 [30,38]. In the past, the reason for using only integer-order models was the lack of methods for solving fractional differential equations. Nowadays, we can find several definitions or approximations methods of fractional order differentiation [4,9,10,13,16,27,28,40,41,49]. Many systems in physics and engineering have been found to be effectively modeled as fractional order systems [2,3,11,20,26,34,37,50,52]. Also, it has been demonstrated that some fractional order differential systems,

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exhibit chaotic behaviors [14,17,29,33,51]. The characterization of chaos in a fractional order dynamical systems has become an increasing need in several applied areas such as robotics, geometric modeling, stability analysis and others. The Lyapunov Characteristic Exponents (LCEs) computations offer good information in understanding the behavior of dynamical systems.

Lyapunov Characteristic Exponents are the average exponential rates of divergence or convergence of nearby orbits in phase space. Measuring the LCE is a fundamental problem in studying any dynamical system because they quantify the exponential divergence of initially close state-space trajectories and estimate the amount of chaos in a system [7]. When a system contains at least one positive Lyapunov exponent it is sufficient for diagnosing chaos and its represents locally instability in a particular direction. The concept of the LCEs has been widely presented in the literature with the description of some theoretical and numerical algorithms for their computation [5,6,18,23,32,39]. The idea introduced by Lyapunov [35] is to evaluate the average growth rate of a small perturbation introduces in the orbit of a dynamical system. Since then, different methods have been developed to calculate the LCE of dynamical systems. Oseledec in [39] was the first to establish a solid theoretical background on calculating the LCE. Then after, Benettin et al. [5,6] developed a complete set of theoretical as well as numerical framework of LCEs. This idea is applied in different systems like infinite-dimensional systems described by delay differential equations, dissipative systems and time series data [43]. When the continuous dynamical system are available, the Jacobian method is well applied, for a system with a time series, the Wolf algorithm [53] is considered, which consists in estimating only the large Lyapunov exponent from time series data after performing a phase-space reconstruction [21].

Other variant methods of LCE like time series based on Wolf algorithm, jacobian method and neural network algorithm method [22,36,53] are associated with difficulties in choosing the embedding dimension and the delay parameter inherent of the phase-space reconstruction [42].

It was observed that the estimating of LES using the classical approach proposed by Wolf et al. in 1985 paper, based on generating a tangent space of the reference (fiducial) system evolution, constructed from the linearized system and subsequent Gram-Schmidt Reorthonormalization (GSR) procedure, doesn't works, since the linearized Fractional Order Dynamical System (FODS) must also be evaluated by pcABM algorithm due to the fractional order nature which takes into account all past variables values, that were previously modified by GSR procedure. An alternative approach to evaluate the LES is to use the Cloned Dynamic method, proposed by Soriano et al. [44]. It is based on time evolution of cloned dynamics with perturbed initial conditions in each variable direction, together with the reference or fiducial system equations.

While the (LCE) was easily computable for integer order systems, the determination of (LCE) of fractional order systems requires the numerical integration of the underlying system. Generally there are three approaches to solve fractional-order differential equations: frequency-domain method [12], Adomian Decomposition Method (ADM) [1] and Adams-Bashforth-Moulton (ABM) algorithm [46]. Computational numeric methods are used to explore FODS. A definition of LCEs for fractional differential systems is introduced in [31] based on frequency-domain approximations. Tavazoei et al. [48] reported that the frequency-domain method is not always reliable in detecting chaos behavior in nonlinear systems. In [8], the LCEs of fractional-order chaotic systems are calculated based on ADM by applying QR decomposition method. The difficult of this approach is the computation of the LCE of the FODS versus the parameter and the order of the system [42]. Recently, Danca et al. [15] investigated the LCE for a Fractional differential equation based on the Benettin-Wolf algorithm using the FDE12.m Matlab program. In fact this program only consider commensurate fractional order systems. Moreover when the fractional order q of the system is low the program is incapable to show a solution. In this work, the predictor-corrector Adams-Bashforth-Moulton (pcABM) algorithm proposed by Diethelm et al. [19] is employed To estimate the LCE of a nonlinear FODS.

The paper is organized as follows, Section 2 presents the predictor-corrector Adams-Bashforth-Moulton (pcABM) algorithm proposed by Diethelm et al. [19] and the cloned dynamical method considered in this paper. Section 3 presents some results and discussions of this method applied in three fractional order chaotic systems, where Jerk system only is also analyzed with classical algorithm as described in [15] and only in commensurate order. 0-1 test [24,25] is also used to reinforce the conclusions about chaotic or non chaotic system dynamics behaviors. Conclusion is carried out in Section 4.

2. Method description

2.1. Improved Adams-Bashforth-Moulton

The algorithm employed for FO systems numerical evaluation is the improved predictor-corrector Adams-Bashforth-Moulton (pcABM) [19] for solution of initial value problem stated by the Volterra Integral equation

$$\begin{cases} {}_C D_t^\alpha x(t) = f(t, x(t)) \\ x(0) = x_0 \end{cases} \quad (1)$$

$$x(t) = x_0 + \frac{1}{\Gamma(\alpha)} \int_0^t (t-\tau)^{\alpha-1} f(\tau, x(\tau)) d\tau \quad (2)$$

The operator ${}_CD_t^\alpha$ denotes the FO derivative in the Caputo sense

$${}_CD_t^\alpha x(t) = \frac{1}{\Gamma(m-\alpha)} \int_0^t \frac{x^m(\tau)}{(t-\tau)^{\alpha+1-m}} d\tau \quad (3)$$

where $\alpha \in \mathbb{R}^+$ is the fractional order, $\Gamma(x)$ is the gamma function and $m = \lceil \alpha \rceil$ is the ceiling function, $m \in \mathbb{N}$ and $m-1 < \alpha \leq m$.

The numerical calculation for $x_h(t_{i+1})$, being $0 \leq t \leq T$ the simulation time interval, $0 \leq i \leq N$ ($N \in \mathbb{Z}^+$) being N the number of steps, and $h = T/N$ the time step:

$$x_h(t_{i+1}) = \sum_{k=0}^{\lceil \alpha \rceil - 1} \left[\frac{t_{i+1}^k}{k!} x_0^{(k)} + \frac{h^\alpha}{\Gamma(\alpha+2)} f(t_{i+1}, x_h^p(t_{i+1})) \right] + \frac{h^\alpha}{\Gamma(\alpha+2)} \sum_{j=0}^n a_{j,i+1} f(t_j, x_h(t_j)) \quad (4)$$

where

$$a_{j,i+1} = \begin{cases} i^{\alpha+1} - (i-\alpha)(i+1)^\alpha, & \text{if } j=0 \\ (i-j+2)^{\alpha+1} + (i-j)^{\alpha+1} - 2(i-j+1)^{\alpha+1}, & \text{if } 1 \leq j \leq i \\ 1, & \text{if } j=i+1 \end{cases} \quad (5)$$

The predictor term is calculated by

$$x_h^p(t_{i+1}) = \sum_{k=0}^{\lceil \alpha \rceil - 1} \frac{t_{i+1}^k}{k!} x_0^{(k)} + \frac{1}{\Gamma(\alpha)} \sum_{j=0}^n b_{j,i+1} f(t_j, x_h(t_j)) \quad (6)$$

where $b_{j,i+1}$ are the predictor coefficients

$$b_{j,i+1} = \frac{h^\alpha}{\alpha} ((i+1-j)^\alpha - (i-j)^\alpha). \quad (7)$$

For the case of n -dimensional Fractional Order Dynamic System (FODS), the trajectory is generated applying the pcABM method with the system equations described by (9) in expanded form

$$\begin{aligned} {}_CD_t^{\alpha_1}(x_1) &= F_1(x_1, t), \\ {}_CD_t^{\alpha_2}(x_2) &= F_2(x_2, t), \\ &\vdots \\ {}_CD_t^{\alpha_n}(x_n) &= F_n(x_n, t), \end{aligned} \quad (8)$$

or in vector form is

$$\begin{cases} {}_CD_t^{\alpha_i} \mathbf{x}(t) = \mathbf{F}(\mathbf{x}, t), \\ \mathbf{x}(t) = [x_1(t), x_2(t), \dots, x_n(t)], \end{cases} \quad (9)$$

where $\mathbf{x}(t)$ are the state vector.

2.2. Computation of Lyapunov exponent spectrum (LES)

The method used for LES estimation is based on cloned dynamic (ClonedDyn) proposed by Soriano et al. [44]. This method evaluate the difference between cloned state vector trajectory and a fiducial vector trajectory. The fiducial evaluation uses initial conditions and the cloned version uses this initial conditions disturbed by small amount δx in orthogonal directions.

To estimate all LES $\lambda_1, \lambda_2, \dots, \lambda_n$, first a large T_{Ref} and N_{Ref} reference system trajectory $\mathbf{x}_{Ref}$ is generated. This attractor operate as rail for LES estimation algorithm to be tracked by the smaller in N and T cloned systems is evaluated with the pcABM method. This system trajectory is generated by eq. (10).

$${}_CD_t^{\alpha_i}(\mathbf{x}_{Ref}) = \mathbf{F}(\mathbf{x}_{Ref}, t), \quad (10)$$

with numerical parameters

$$\begin{cases} 0 \leq t \leq T_{Ref}, \\ 0 \leq n \leq N_{Ref} \quad (N_{Ref} \in \mathbb{Z}^+), \\ h = T_{Ref}/N_{Ref}, \end{cases} \quad (11)$$

where T_{Ref} is the time interval, N_{Ref} is the number of points within this interval and h is the time step for the reference system.

The cloned systems are numerically evaluated applying the pcABM method on the system $n + 1$ equations, n for cloned systems ($\mathbf{x}_{C1} \dots \mathbf{x}_{Cn}$) and 1 for a fiducial system $\mathbf{x}_{C0}$ based on equations as follows

$$\begin{aligned} {}_C D_t^{\alpha_i}(\mathbf{x}_{C0}) &= \mathbf{F}(\mathbf{x}_{C0}, t), \\ {}_C D_t^{\alpha_i}(\mathbf{x}_{C1}) &= \mathbf{F}(\mathbf{x}_{C1}, t), \\ {}_C D_t^{\alpha_i}(\mathbf{x}_{C2}) &= \mathbf{F}(\mathbf{x}_{C2}, t), \\ &\vdots \\ {}_C D_t^{\alpha_i}(\mathbf{x}_{Cn}) &= \mathbf{F}(\mathbf{x}_{Cn}, t). \end{aligned} \quad (12)$$

These simulations are made with T_{Clone} time interval, N_{Clone} number of points within this interval and clone time step hC . The clones are evaluated $K = T_{Ref}/T_{Clone}$ times.

$$\begin{cases} 0 \leq t \leq T_{Clone}, \\ 0 \leq n \leq N_{Clone} \quad (N_{Clone} \in \mathbb{Z}^+), \\ hC = T_{Clone}/N_{Clone}. \end{cases} \quad (13)$$

For example, a reference system generated with $T_{Ref} = 1000$ s and $N_{Ref} = 100000$ points, resulting $h = T_{Ref}/N_{Ref} = 0.01$ as time step. This reference trajectory can be implemented using 10 chained blocks of $T = 100$ s and $N = 10000$ for time saving. LES evaluation propose with this reference system can be done without great difference as if it was generated at once. The LES evaluation can be made with clone simulations with $T_{Clone} = 10$ s, $N_{Clone} = 1000$, $hC = 0.01$ and $K = T_{Ref}/T_{Clone} = 100$.

The fiducial system ($\mathbf{x}_{C0}$) receives the $\mathbf{x}_0$ initial conditions and the cloned versions, the $\mathbf{x}_0$ initial conditions but disturbed in orthogonal direction as follows

$$\begin{aligned} \mathbf{x}_0^{(0)} &= [x_1(0), x_2(0), \dots, x_n(0)], \\ x_{0C0}^{(0)} &= \mathbf{x}_0^{(0)}, \\ x_{0C1}^{(0)} &= \mathbf{x}_0^{(0)} + \delta x \cdot \mathbf{e}_1, \\ x_{0C2}^{(0)} &= \mathbf{x}_0^{(0)} + \delta x \cdot \mathbf{e}_2, \\ &\vdots \\ \mathbf{x}_{0Cn}^{(0)} &= \mathbf{x}_0^{(0)} + \delta x \cdot \mathbf{e}_n, \end{aligned} \quad (14)$$

where

$$\begin{aligned} \mathbf{e}_1 &= [1, 0, \dots, 0], \\ \mathbf{e}_2 &= [0, 1, \dots, 0], \\ &\vdots \\ \mathbf{e}_n &= [0, 0, \dots, 1]. \end{aligned} \quad (15)$$

After simulation time interval T_{Clone} (clone simulation time interval), the difference vectors between the fiducial system and cloned systems are calculated

$$\begin{aligned} \delta \mathbf{v}_1^{(1)} &= \mathbf{x}_{C1}(T_C) - \mathbf{x}_{C0}(T_C), \\ \delta \mathbf{v}_2^{(1)} &= \mathbf{x}_{C2}(T_C) - \mathbf{x}_{C0}(T_C), \\ &\vdots \\ \delta \mathbf{v}_n^{(1)} &= \mathbf{x}_{Cn}(T_C) - \mathbf{x}_{C0}(T_C). \end{aligned} \quad (16)$$

To avoid numerical errors because the fact that the difference vectors $\delta \mathbf{v}_1, \dots, \delta \mathbf{v}_n$ tend to align to the most expansive direction with the dynamical system evolution, the Gram-Schmidt Reorthonormalization procedure is employed as follows.

Defining Inner product as:

$$\langle \mathbf{v}, \mathbf{w} \rangle = \sum_{k=1}^n v_k \cdot w_k = \text{proj}_{\mathbf{v}}(\mathbf{w}) \quad (17)$$

and vector norm as:

$$\|\mathbf{v}\| = \sum_{k=1}^n v_k^2 \quad (18)$$

The reorthonormalizations is computed as follows

$$\begin{aligned} \mathbf{v}_1^{(1)} &= \delta \mathbf{v}_1^{(1)} \\ \mathbf{u}_1^{(1)} &= \frac{\mathbf{v}_1^{(1)}}{\|\mathbf{v}_1^{(1)}\|}, \\ \mathbf{v}_2^{(1)} &= \delta \mathbf{v}_2^{(1)} - \langle \delta \mathbf{v}_2^{(1)}, \mathbf{u}_1^{(1)} \rangle \mathbf{u}_1^{(1)}, \\ \mathbf{u}_2^{(1)} &= \frac{\mathbf{v}_2^{(1)}}{\|\mathbf{v}_2^{(1)}\|} \\ &\vdots \\ \mathbf{v}_n^{(1)} &= \delta \mathbf{v}_n^{(1)} - \langle \delta \mathbf{v}_n^{(1)}, \mathbf{u}_1^{(1)} \rangle \mathbf{u}_1^{(1)} - \langle \delta \mathbf{v}_n^{(1)}, \mathbf{u}_2^{(1)} \rangle \mathbf{u}_2^{(1)} - \dots \\ &\quad - \langle \delta \mathbf{v}_n^{(1)}, \mathbf{u}_{n-1}^{(1)} \rangle \mathbf{u}_{n-1}^{(1)}. \end{aligned} \quad (19)$$

The new initial conditions are set from the reference system as described in (14) and another T_C interval is simulated. The fiducial system receives the N_C position from reference system $\mathbf{x}_{\text{Ref}}$ as initial conditions and the cloned systems, these coordinates with the δ_x displacement in each normalized direction $\mathbf{x}_{0Cn}^{(1)} = \mathbf{x}_0^{(1)} + \delta_x \cdot \mathbf{u}_n^{(1)}$ as follows

$$\begin{aligned} \mathbf{x}_0^{(1)} &= [x_{\text{Ref}1}(N_C), x_{\text{Ref}2}(N_C), \dots, x_{\text{Ref}n}(N_C)], \\ \mathbf{x}_{0C0}^{(1)} &= \mathbf{x}_0^{(1)}, \\ \mathbf{x}_{0C1}^{(1)} &= \mathbf{x}_0^{(1)} + \delta_x \cdot \mathbf{u}_1^{(1)}, \\ \mathbf{x}_{0C2}^{(1)} &= \mathbf{x}_0^{(1)} + \delta_x \cdot \mathbf{u}_2^{(1)}, \\ &\vdots \\ \mathbf{x}_{0Cn}^{(1)} &= \mathbf{x}_0^{(1)} + \delta_x \cdot \mathbf{u}_n^{(1)}. \end{aligned} \quad (20)$$

This procedure is repeated K times leading to the LE spectrum, large enough to assure LES convergence

$$\lambda_n = \lim_{\delta_x \rightarrow 0} \lim_{K \rightarrow \infty} \frac{1}{K \cdot T_{\text{Clone}}} \sum_{k=1}^K \ln \left\| \frac{\mathbf{v}_n^{(k)}}{\delta_x} \right\|. \quad (21)$$

Evaluated by (21) with cloned systems initial conditions taken from reference system $\mathbf{x}^{(k)}(0) = [x_{\text{Ref}1}(k \cdot N_C), x_{\text{Ref}2}(k \cdot N_C), \dots, x_{\text{Ref}n}(k \cdot N_C)] = \mathbf{x}_0^{(k)}$.

Precaution must be taken in choosing values of T_C and K because the simulations must explore the dynamical evolution in a long range. But if T is large, the LE calculations grows to large values and estimation errors appear. A plotting of results must reveal the K value that results in a satisfactory evaluation of LE spectrum.

3. Results and discussion

3.1. Application in a Jerk System

A Jerk System is one that appears a third-order derivative, corresponding to time derivative of acceleration in mechanical systems. Eq. (22) describes the Jerk system studied and proposed by Sprott [45] that will be studied, in integer order version

Table 1

Simulation parameters for Jerk System LES estimation using pcABM-ClonedDyn method.

Condition	α_1	α_2	α_3
Commensurate I	1.0	1.0	1.0
Commensurate II	0.9	0.9	0.9
Commensurate III	0.8	0.8	0.8
Commensurate IV	0.7	0.7	0.7
Incommensurate I	0.9	1.0	1.0
Incommensurate II	0.8	1.0	1.0
Incommensurate III	0.7	1.0	1.0
Incommensurate IV	0.6	1.0	1.0

Table 2

The clone simulations time and step parameters for Jerk System LES estimation.

Condition	T_Clone (s)	N_Clone	hC (s)	K
I	1	100	0.01	1000
II	5	500	0.01	200
III	10	1000	0.01	100

$$\ddot{x} + a.\ddot{x} + f(\dot{x}) + x = 0. \quad (22)$$

This system is represented in state space form as follows

$$\begin{cases} \frac{dx}{dt} = y, \\ \frac{dy}{dt} = z, \\ \frac{dz}{dt} = -a.z - f(y) - x, \end{cases} \quad (23)$$

and in fractional order derivatives case

$$\begin{cases} {}^C D_t^{\alpha_1}(x) = y, \\ {}^C D_t^{\alpha_2}(y) = z, \\ {}^C D_t^{\alpha_3}(z) = -a.z - f(y) - x. \end{cases} \quad (24)$$

The non-linear function $f(y)$ shown in (24) describes an electronic diode model for characteristic curve, with $I_c = 10.10^{-9}$ A, $V_T = 0.026$ V and n is a characteristic parameter ($n = 2$ in this work)

$$f(y) = I_c \cdot [\exp(y/nV_T) - 1]. \quad (25)$$

The initial conditions for simulations were $x_0 = [0.1, 0.1, 0.1]$. The total reference system simulation time is $T_{Ref} = 1000$ s, with $N = 100000$ steps and time step $h = 0.01$ s and the $a = 0.5$ for system parameter, resulting chaotic behavior [45].

Eight derivative orders configurations are explored being four for each commensurate and incommensurate as described in Table 1. The clone simulations time T_Clone , step number N_Clone , time step hC and total number of LES steps K are presented in Table 2. The perturbation in orthonormal directions is $\delta \mathbf{x}_0 = 0.001$.

The Fig. 1(a) phase plot show clearly a chaotic behavior as a strange attractor for the commensurate order $\alpha_1 = \alpha_2 = \alpha_3 = 1.0$, which correspond in $(+, 0, -)$ configuration for $(\lambda_1, \lambda_2, \lambda_3)$. The Figs. 1(b), (c) and (d) shows the LES estimation as function of K and for different T_Clone , with stable and convergent result. Despite the slightly positive value for λ_2 , all this results can be considered satisfactory to characterize the chaotic behavior for this commensurate configuration.

Fig. 1 (e) shows the commensurate order $\alpha_1 = \alpha_2 = \alpha_3 = 0.9$ phase plot denoting a limit cycle attractor. For this behavior, the LES $(\lambda_1, \lambda_2, \lambda_3)$ must result in $(0, 0, -)$ or $(0, -, -)$ configuration, which is confirmed with the result shown in Fig. 1 (g) and (h) for $T_Clone = 10$ s and 5 s respectively, with $(0, 0, -)$ configuration. The results for $T_Clone = 1$, Fig. 1 (f), denotes chaotic $(+, -, -)$ LES result, what doesn't correspond to phase plot, showing the strong dependence of LES estimation on T_Clone .

The phase plot for commensurate order $\alpha_1 = \alpha_2 = \alpha_3 = 0.8$ is shown in Fig. 1 (i), and a limit cycle attractor. The LES estimation results are $(+, -, -)$ for $T_Clone = 1$ s, $(-, -, -)$ for 5 s and $(0, -, -)$ for 10 s, being the 10 s the more representative result.

Another interesting behavior was observe with commensurate order $\alpha_1 = \alpha_2 = \alpha_3 = 0.7$ phase plot, resulting in a stable point as shown in Fig. 1 (m). The results for LES estimation shown in Fig. 1 (n) and (o), corresponding to $T_Clone = 1$ s

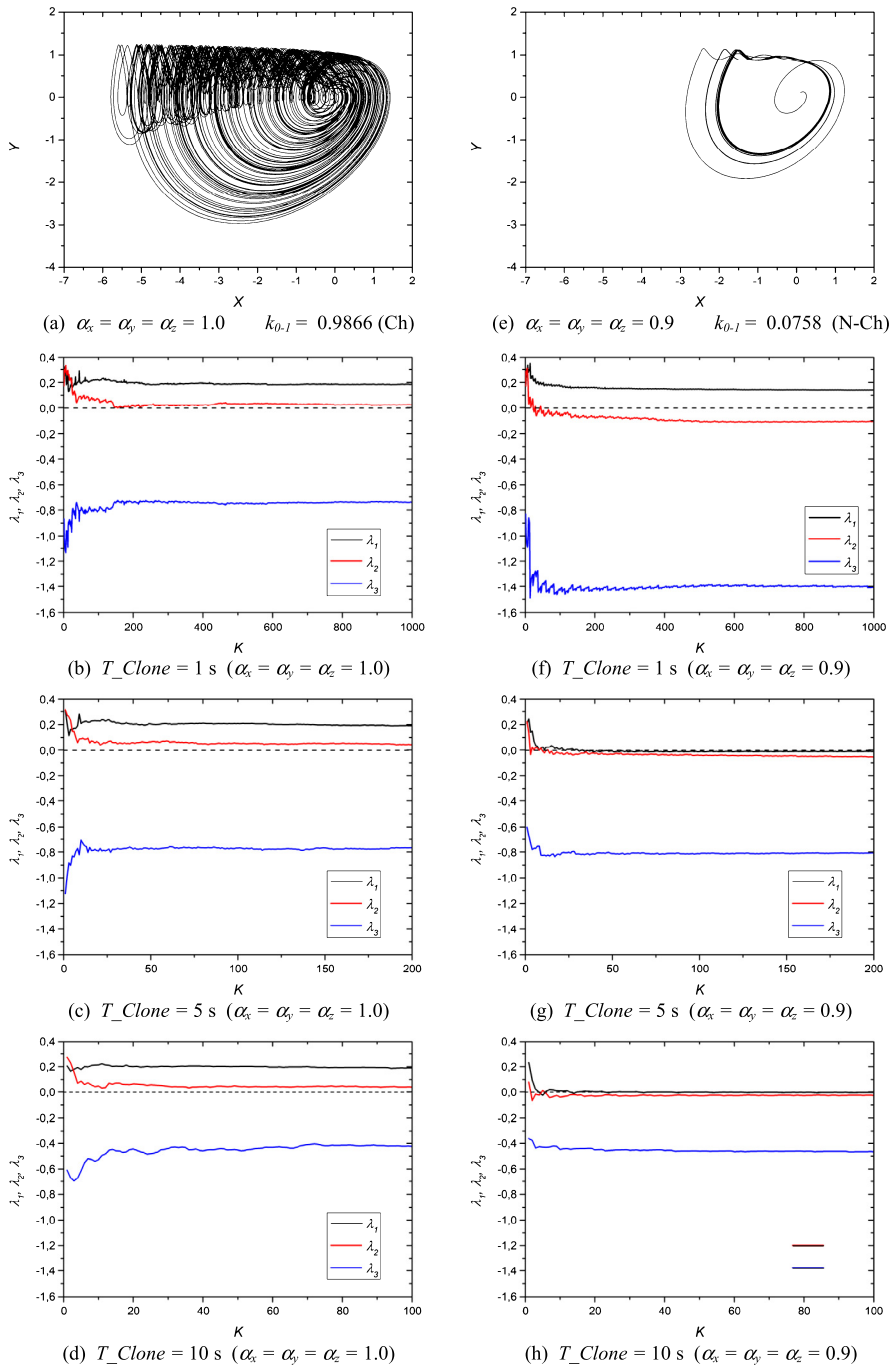


Fig. 1. Jerk System LES results. (a) Commensurate I order $\alpha_x = \alpha_y = \alpha_z = 1.0$ phase plot; LES results for $T_Clone = 1$ s (b), $T_Clone = 5$ s (c), $T_Clone = 10$ s (d); (e) Commensurate II order $\alpha_x = \alpha_y = \alpha_z = 0.9$ phase plot; LES results for $T_Clone = 1$ s (f), $T_Clone = 5$ s (g), $T_Clone = 10$ s (h); (i) Commensurate III order $\alpha_x = \alpha_y = \alpha_z = 0.8$ phase plot; LES results for $T_Clone = 1$ s (j), $T_Clone = 5$ s (k), $T_Clone = 10$ s (l); (m) Commensurate IV order $\alpha_x = \alpha_y = \alpha_z = 0.7$ phase plot; LES results for $T_Clone = 1$ s (n), $T_Clone = 5$ s (o), $T_Clone = 10$ s (p). Jerk System LES results using Classical Method, Commensurate order phase plot for (r) $\alpha_x = \alpha_y = \alpha_z = 1.0$, (s) $\alpha_x = \alpha_y = \alpha_z = 0.9$, (t) $\alpha_x = \alpha_y = \alpha_z = 0.8$, (u) $\alpha_x = \alpha_y = \alpha_z = 0.7$. LES results for (v) $\alpha_x = \alpha_y = \alpha_z = 1.0$, (w) $\alpha_x = \alpha_y = \alpha_z = 1.0$, (x) $\alpha_x = \alpha_y = \alpha_z = 1.0$ and (y) $\alpha_x = \alpha_y = \alpha_z = 1.0$. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

and 5 s respectively, are clearly inconsistent. $T_Clone = 1$ s result in $(+, +, -)$ configuration, a typical result for chaos and for 5 s, the $(0, 0, -)$ configuration which corresponds to a limit cycle type of attractor. The LES estimation with $T_Clone = 10$ s resulted in $(-, -, -)$ configuration, corresponding to stable point. These results shows that care must be taken when

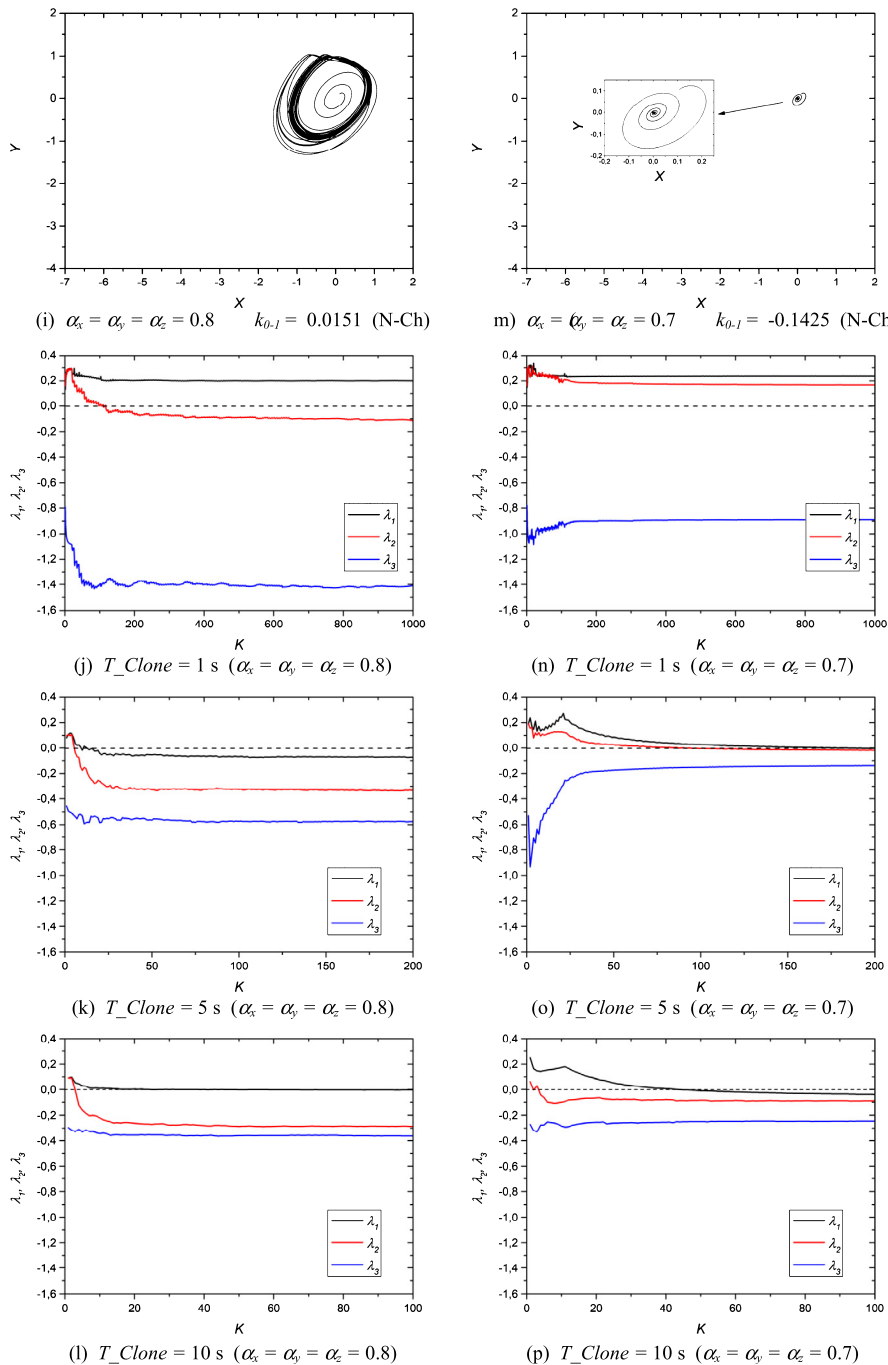


Fig. 1. continued

choosing T_{Clone} values, this choice impacting directly in K determination as can be seen in Table 2, which value would theoretically trend to infinity, eq. (21).

To compare with Classical Method where tangent trajectories are analyzed, the system Jacobian is needed [22]. Results using MATLAB code found in [15] for commensurate systems are presented and named as Classical Method LCE results, shown in Fig. 1 (r) to (y). The numerical simulation parameters are $h = 0.005$ for simulation time step, $h_{norm} = 5$ for fractional order number of steps, $t_{end} = 5000$ s and $x_{start} = [0.1, 0.1, 0.1]$ as initial conditions. Table 3 shows these results.

The incommensurate analysis are shown in Fig. 2. Only α_x is varied with values of 0.9, 0.8, 0.7 and 0.6. The phase plots for $\alpha_x = 0.9$ and 0.8 resulted in chaotic behavior, as can be seen in Fig. 2 (a) and (e), and from LES estimations shown in

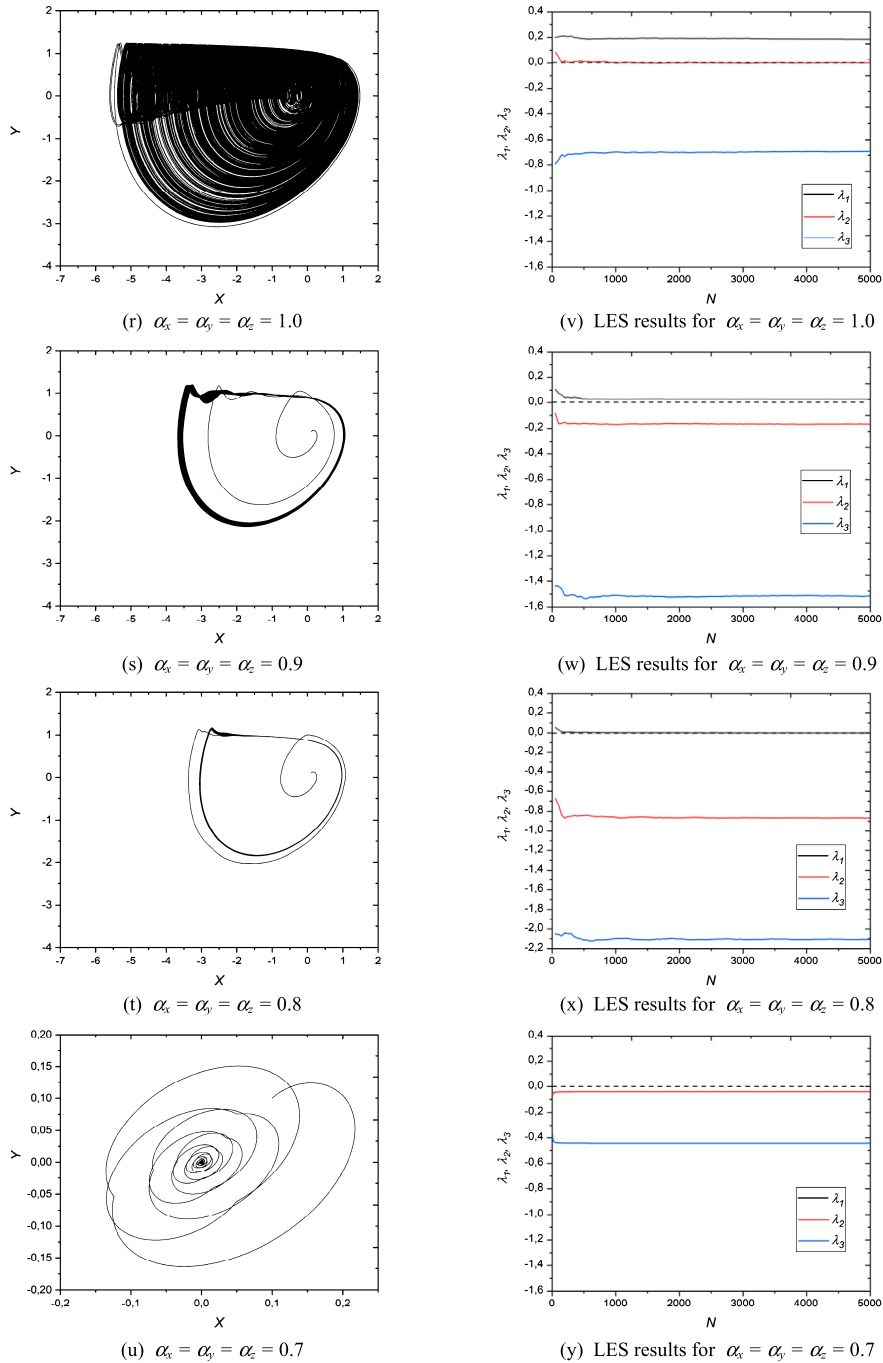


Fig. 1. continued

Fig. 2 (d) and (h), both for $T_{Clone} = 10$ s. The case of $\alpha_x = 0.9$, the LES estimations with $T_{Clone} = 10$ s resulted more realistic than with $T_{Clone} = 1$ s and 5 s considering the $(+, 0, -)$ configuration for $(\lambda_1, \lambda_2, \lambda_3)$. For the case of $\alpha_x = 0.8$, the result for $T_{Clone} = 1$ s shown in Fig. 2 (i) which has the $(+, +, -)$ configuration, being results shown in Fig. 2 (g) and (h) for $T_{Clone} = 5$ s and 10 s more representative for chaotic LES results with $(+, 0, -)$ configuration.

For the cases of $\alpha_x = 0.7$ and 0.6, the resulting phase plot shown in Fig. 2 (i) and (m) and LES estimation for $T_{Clone} = 10$ s signs that are limit cycle attractor cases. For $\alpha_x = 0.7$, the LES estimation with $T_{Clone} = 1$ s, Fig. 2 (j), reveals a chaotic behavior. With $T_{Clone} = 5$ s, Fig. 2 (k), a limit cycle attractor with $(+, +, -)$ LES configuration and with $T_{Clone} = 10$ s, a limit cycle attractor with $(+, 0, -)$ LES configuration. For $\alpha_x = 0.6$, both $T_{Clone} = 1$ and 5 s, Fig. 2 (n)

Table 3
Jerk System - LCE asymptotic values: Cloned Dynamics x Classical.

α_1	α_2	α_3		λ_1	λ_2	λ_3
1.0	1.0	1.0	Cloned Dynamic	0.1899	0.0413	-0.4246
			Classic	0.1845	-0.0003	-0.6930
0.9	0.9	0.9	Cloned Dynamic	-0.0042	-0.0267	-0.4656
			Classic	-0.0243	-0.1376	-1.3396
0.8	0.8	0.8	Cloned Dynamic	-0.0047	-0.2864	-0.3634
			Classic	0.0005	-0.8676	-2.1037
0.7	0.7	0.7	Cloned Dynamic	-0.0527	-0.0528	-0.2763
			Classic	0.0006	-0.6493	-2.1408

and (o), shows chaotic LES results with (+ + −) and (+ − −) configurations respectively, when with $T_{Clone} = 10$ s, Fig. 2 (p), the limit cycle is confirmed with (0, −, −) LES configuration.

The analysis of results shows clearly the strong dependence on T_{Clone} for LES estimation. For the Jerk system studied, T_{Clone} for the more accurate LES evaluation is with $T_{Clone} = 10$ s.

3.2. Application in a Financial System

The Financial System studied is based on Tacha et al. work [47] where x represents the interest rate, y the investment demand and z the price index. The equations in integer order form are as follows

$$\begin{cases} \frac{dx}{dt} = z + (y - \alpha)x, \\ \frac{dy}{dt} = 1 - by - |x|, \\ \frac{dz}{dt} = -x - cz, \end{cases} \quad (26)$$

and in fractional order derivatives case are

$$\begin{cases} {}^C D_t^{\alpha_1}(x) = z + (y - a)x, \\ {}^C D_t^{\alpha_2}(y) = 1 - by - |x|, \\ {}^C D_t^{\alpha_3}(z) = -x - cz, \end{cases} \quad (27)$$

with parameters $a = 1.0$, $b = 0.15$ and $c = 1.0$, corresponding in chaotic behavior in integer order. This system has a non-linearity type modulus function that causes a derivative indetermination at $x = 0$ what turns impossible to determine the Jacobian for LES estimation using tangent maps algorithms. In this case it is possible to employ the Cloned Dynamics to estimate the LES.

The initial conditions for simulations were $x_0 = [1.0, 1.0, 1.0]$. The total reference system simulation time is $T_{Ref} = 5000$ s, with $N = 100000$ steps and time step $h = 0.05$ s. The derivative order configurations are the same as explored in Jerk System being four for each commensurate and incommensurate as described in Table 1. The clone simulations conditions are: total time $T_{Clone} = 10$ s, step number $N_{Clone} = 1000$, time step $hC = 0.01$ s and total number of LES steps $K = 500$. The perturbation in orthonormal directions is $\delta \mathbf{x}_0 = 0.001$.

As can be observed in Fig. 3, all results are consistent. The cases commensurate for $\alpha_x = \alpha_y = \alpha_z = 1.0$ and 0.9 are clearly chaotic, resulting in LES configuration type (+, 0, −), and resulting in stable fixed point for $\alpha_x = \alpha_y = \alpha_z = 0.8$ and 0.7, which results in (−, −, −) LES configuration.

For the incommensurate cases, $\alpha_y = \alpha_z = 1.0$, $\alpha_x = 0.9$, 0.8 and 0.7, the results are chaotic with (+, 0, −) LES configuration and $\alpha_x = 0.7$ resulted in stable fixed point and (−, −, −) LES configuration, as illustrated in Fig. 4.

3.3. Application in a Four Wing Chaotic System

A third and interesting system that presents chaotic behavior is analyzed. This system is derived from Euler motion equations and its details is described in [54]:

$$\begin{cases} \frac{dx}{dt} = -x + yz, \\ \frac{dy}{dt} = y - xz, \\ \frac{dz}{dt} = b - cz + xy, \end{cases} \quad (28)$$

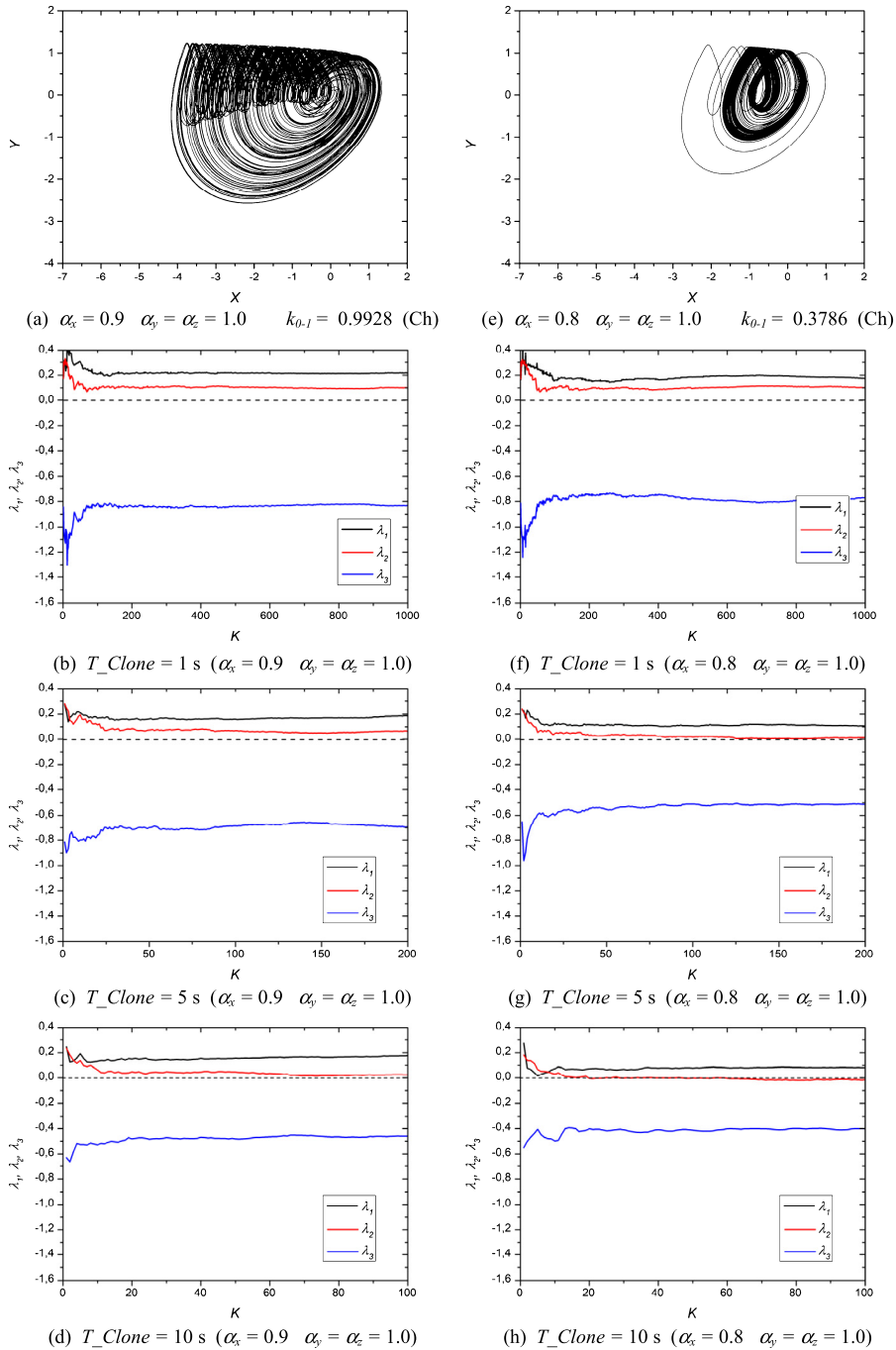


Fig. 2. Jerk System LES results. (a) Incommensurate I order $\alpha_x = 0.9 \alpha_y = \alpha_z = 1.0$ phase plot; LES results for $T_{\text{Clone}} = 1 \text{ s}$ (b), $T_{\text{Clone}} = 5 \text{ s}$ (c), $T_{\text{Clone}} = 10 \text{ s}$ (d); (e) Incommensurate II order $\alpha_x = 0.8 \alpha_y = \alpha_z = 1.0$ phase plot; LES results for $T_{\text{Clone}} = 1 \text{ s}$ (f), $T_{\text{Clone}} = 5 \text{ s}$ (g), $T_{\text{Clone}} = 10 \text{ s}$ (h); (i) Incommensurate III order $\alpha_x = 0.7 \alpha_y = \alpha_z = 1.0$ phase plot; LES results for $T_{\text{Clone}} = 1 \text{ s}$ (j), $T_{\text{Clone}} = 5 \text{ s}$ (k), $T_{\text{Clone}} = 10 \text{ s}$ (l); (m) Incommensurate IV order $\alpha_x = 0.6 \alpha_y = \alpha_z = 1.0$ phase plot; LES results for $T_{\text{Clone}} = 1 \text{ s}$ (n), $T_{\text{Clone}} = 5 \text{ s}$ (o), $T_{\text{Clone}} = 10 \text{ s}$ (p).

and in fractional order derivatives case are

$$\begin{cases} {}^C D_t^{\alpha_1}(x) = -x + yz, \\ {}^C D_t^{\alpha_2}(y) = y - xz, \\ {}^C D_t^{\alpha_3}(z) = b - cz + xy. \end{cases} \quad (29)$$

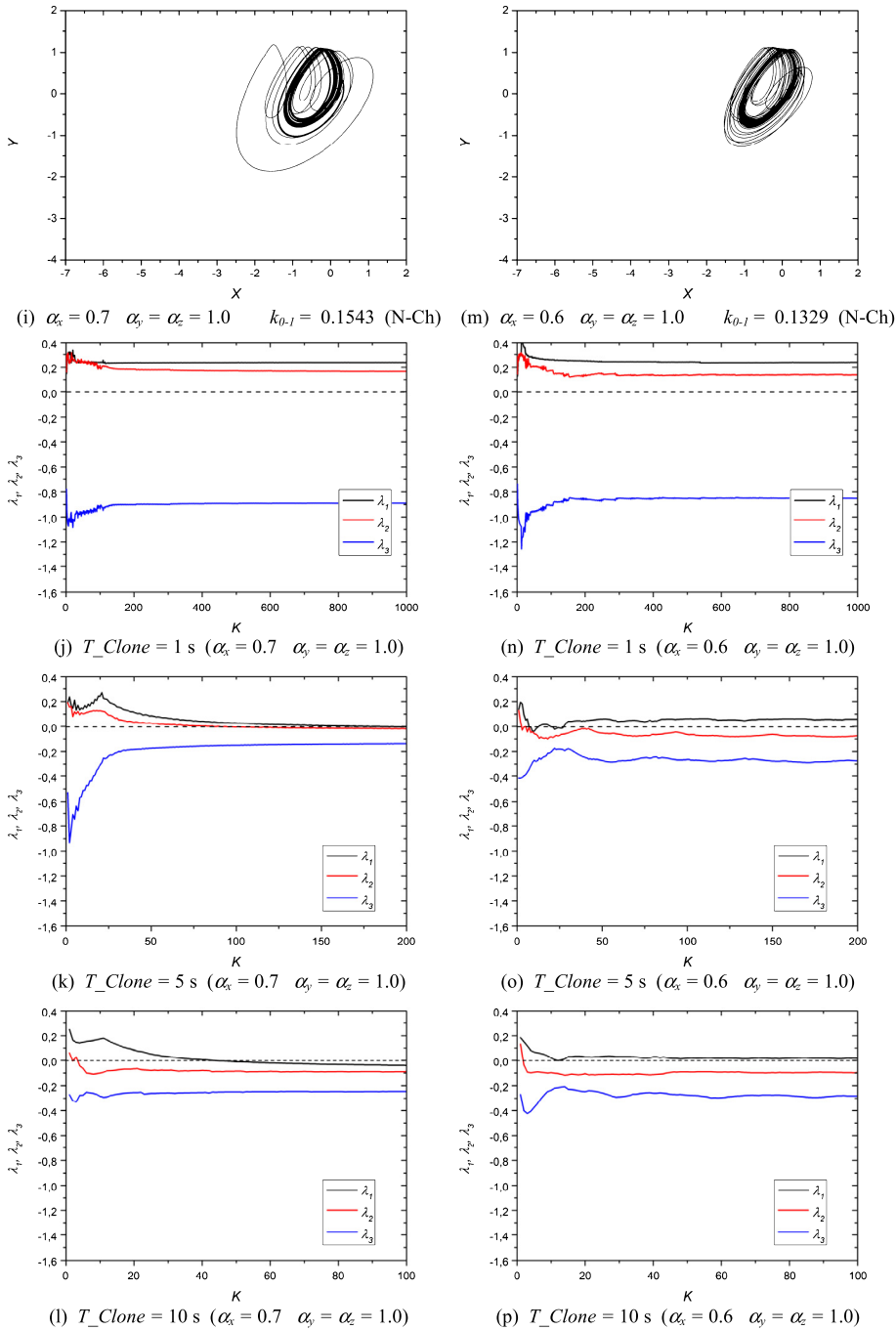


Fig. 2. continued

The system of equations (28) is a three dimensional quadratic autonomous system with x, y, z the states variables. The system is the generalized two wing equation, it possesses three nonlinear term yz, xz, yx , and two positive real parameters b, c . This system is found to be chaotic for $b = 0.53$ and $c = 3.0$.

The initial conditions for simulations were $x_0 = [0.1, 0.1, 0.1]$. The total reference system simulation time is $T_{Ref} = 1000$ s, with $N = 100000$ steps and time step $h = 0.01$ s.

The derivative order configurations are the same as explored in Jerk System being four for each commensurate and incommensurate as described in Table 1. The clone simulations conditions are: total time $T_{Clone} = 5$ s, step number

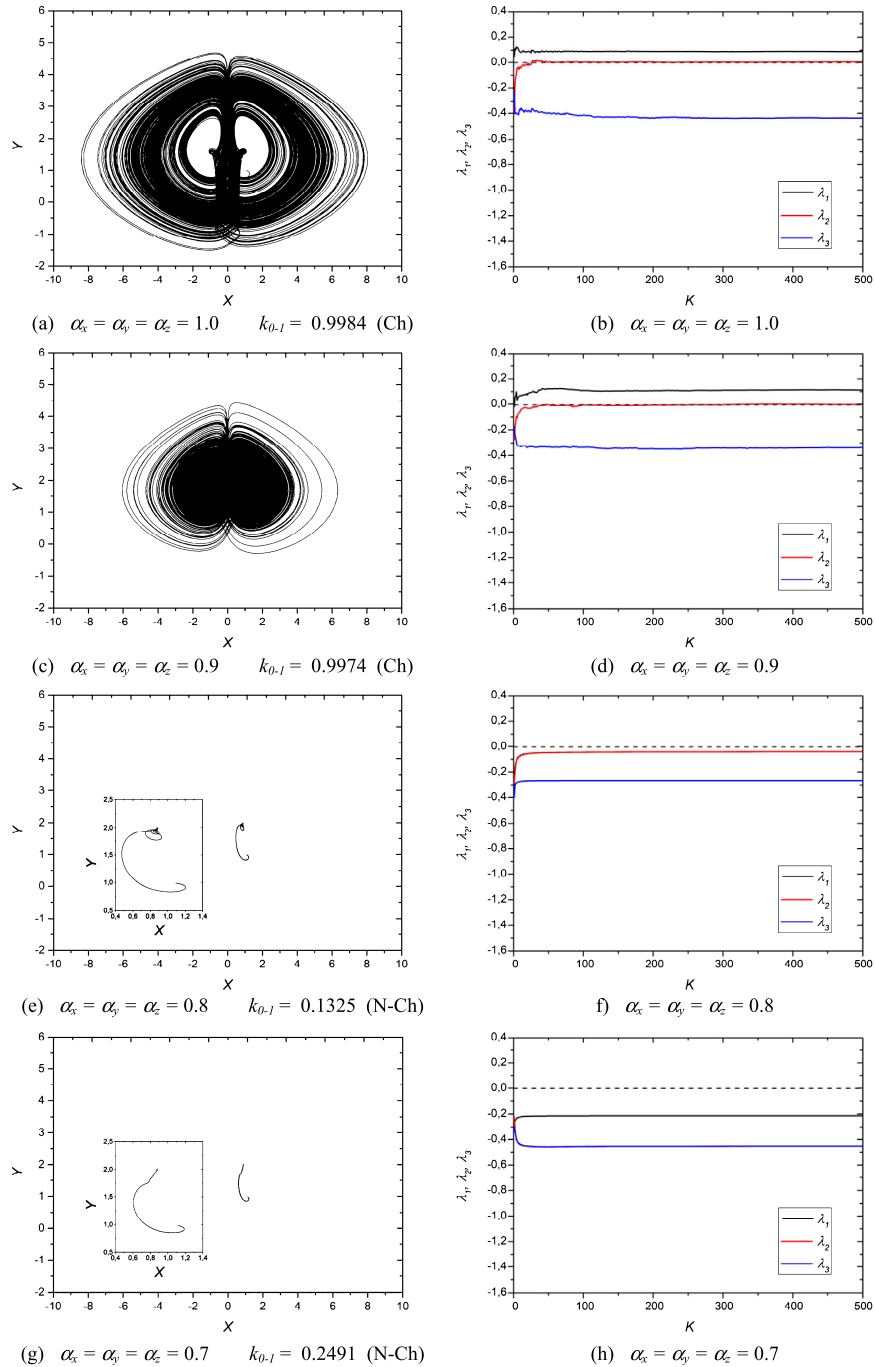


Fig. 3. Financial System LES results. (a) Commensurate order $\alpha_x = \alpha_y = \alpha_z = 1.0$ phase plot and (b) LES; (c) Commensurate order $\alpha_x = \alpha_y = \alpha_z = 0.9$ phase plot and (d) LES; (e) Commensurate order $\alpha_x = \alpha_y = \alpha_z = 0.8$ phase plot and (f) LES; (g) Commensurate order $\alpha_x = \alpha_y = \alpha_z = 0.7$ phase plot and (h) LES.

$N_{Clone} = 500$, time step $hC = 0.01$ s and total number of LES steps $K = 200$. The perturbation in orthonormal directions is $\delta \mathbf{x}_0 = 0.001$.

The simulation results are shown in Fig. 5 and 6 for commensurate and incommensurate cases respectively, and LCE are consistent with phase plane results. The cases commensurate for $\alpha_x = \alpha_y = \alpha_z = 1.0$ and 0.9 are clearly chaotic, resulting in LES configuration type $(+, 0, -)$, and resulting in stable fixed point for $\alpha_x = \alpha_y = \alpha_z = 0.8$ and 0.7 , which results in $(-, -, -)$ LES configuration.

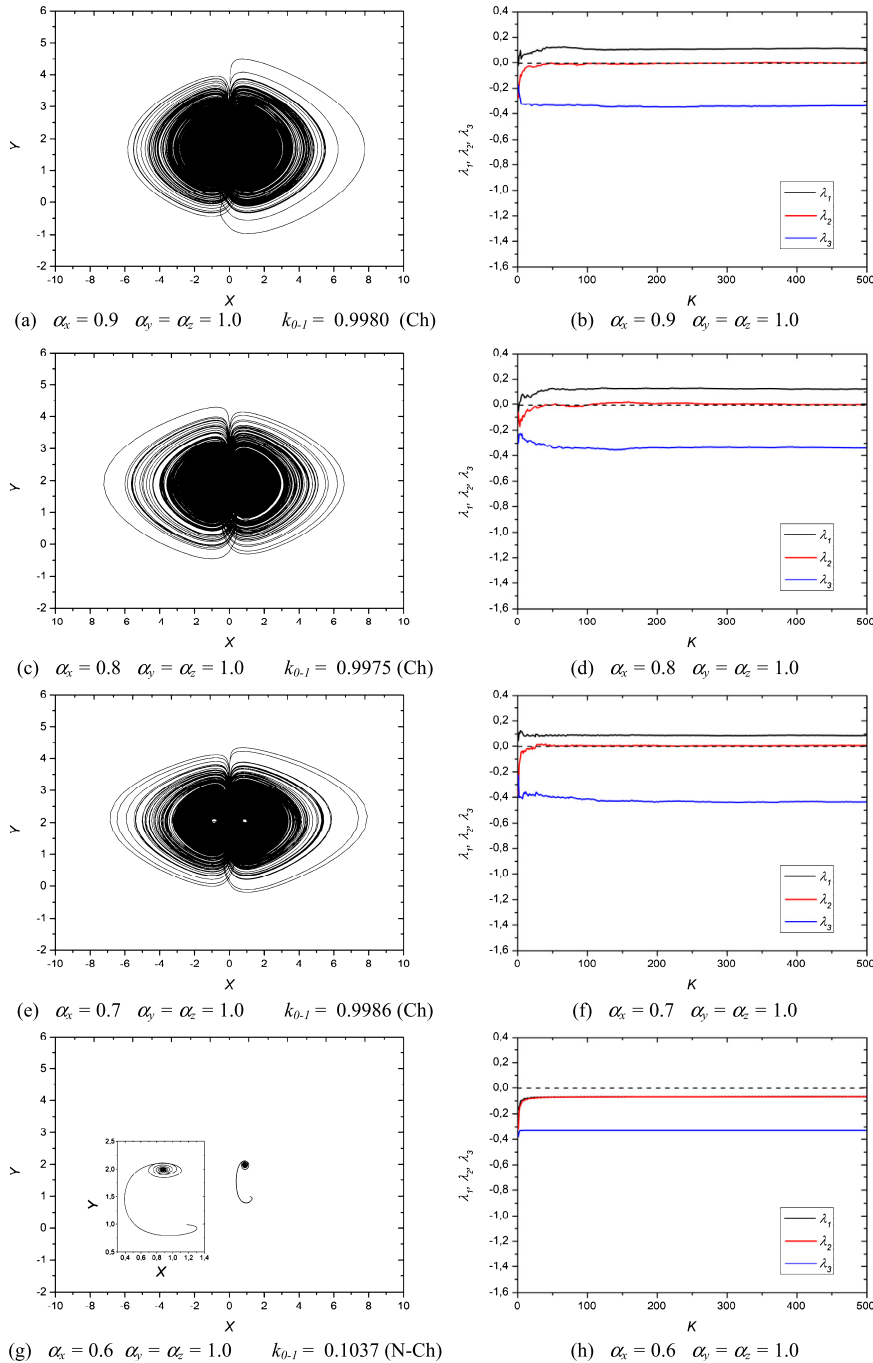


Fig. 4. Financial System LES results. (a) Incommensurate order $\alpha_x = 0.9\alpha_y = \alpha_z = 1.0$ phase plot and (b) LES; (c) Incommensurate order $\alpha_x = 0.8\alpha_y = \alpha_z = 1.0$ phase plot and (d) LES; (e) Incommensurate order $\alpha_x = 0.7\alpha_y = \alpha_z = 1.0$ phase plot and (f) LES; (g) Incommensurate order $\alpha_x = 0.6\alpha_y = \alpha_z = 1.0$ phase plot and (h) LES.

For the incommensurate cases, $\alpha_y = \alpha_z = 1.0\alpha_x = 0.9, 0.8, 0.7$ and 0.6 , the results are chaotic with $(+, 0, -)$ LES configuration, in agreement with phase plane results.

3.4. Application of 0-1 test to confirm chaotic behavior

The 0-1 Test for Chaos method is employed to add information and confirm chaotic or non-chaotic system behavior. The results are presented in Tables 4, 5 and 6, where the LEC asymptotic values are presented for the three systems studied in eight situations for derivative orders, together with the 0-1 Test of Chaos k_{0-1} values. The routine employed uses correlation

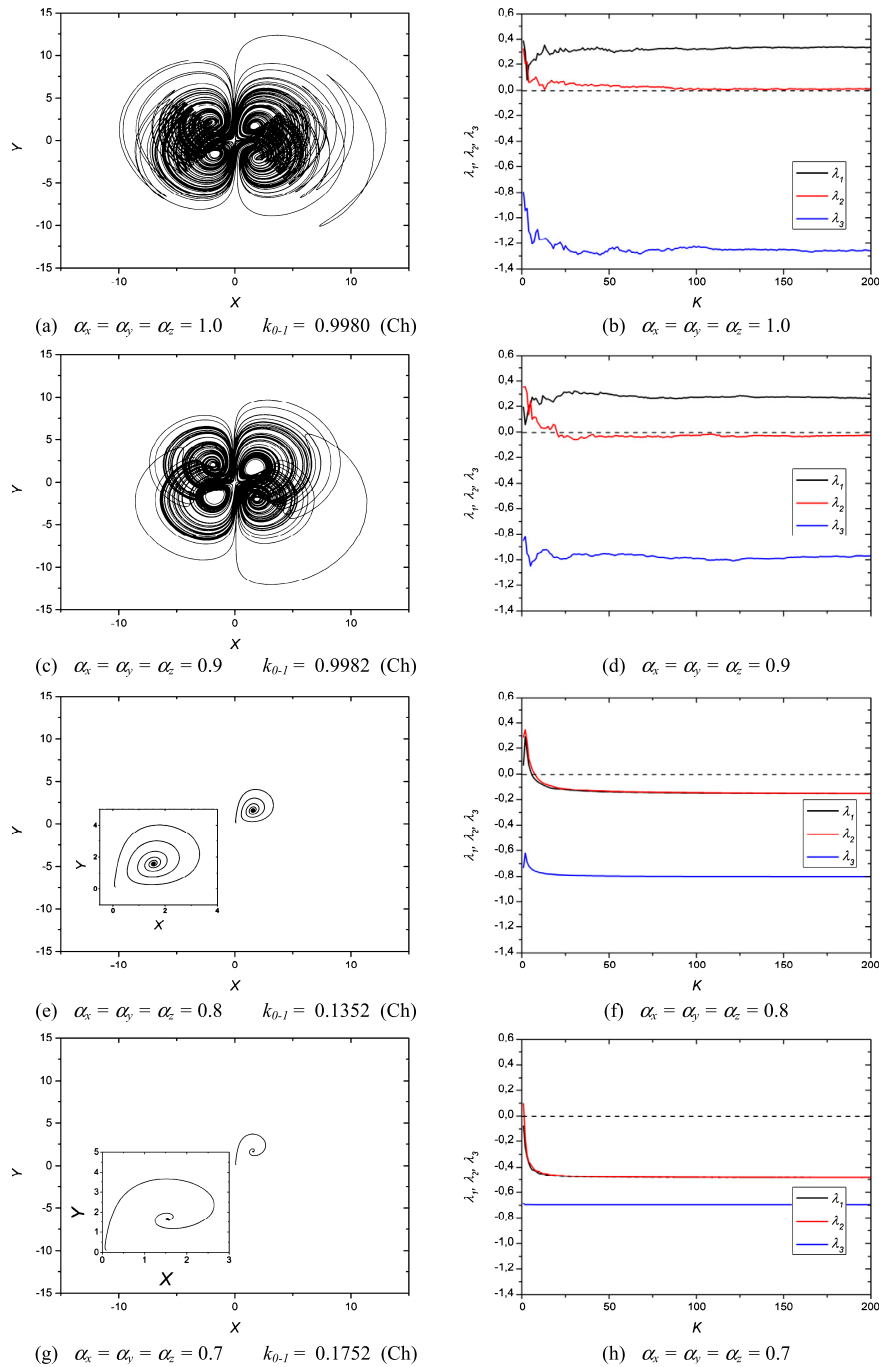


Fig. 5. Four Wing Chaotic System LES results. (a) Commensurate order $\alpha_x = \alpha_y = \alpha_z = 1.0$ phase plot and (b) LES; (c) Commensurate order $\alpha_x = \alpha_y = \alpha_z = 0.9$ phase plot and (d) LES; (e) Commensurate order $\alpha_x = \alpha_y = \alpha_z = 0.8$ phase plot and (f) LES; (g) Commensurate order $\alpha_x = \alpha_y = \alpha_z = 0.7$ phase plot and (h) LES.

function for the k_{0-1} evaluation, and are presented and discussed in [24,25]. The chaotic behavior, which corresponds to values of k_{0-1} very near 1, are shown in bold face to highlight from non-chaotic behavior, which are values near 0, despite some intermediate values. The k_{0-1} results are also presented at phase plot figures captions.

4. Conclusion

A method for Lyapunov Characteristic Exponents based on Coned Dynamics is proposed to evaluate chaotic behavior in Fractional order dynamic systems as an alternative method for evaluation, supporting both incommensurate fractional order

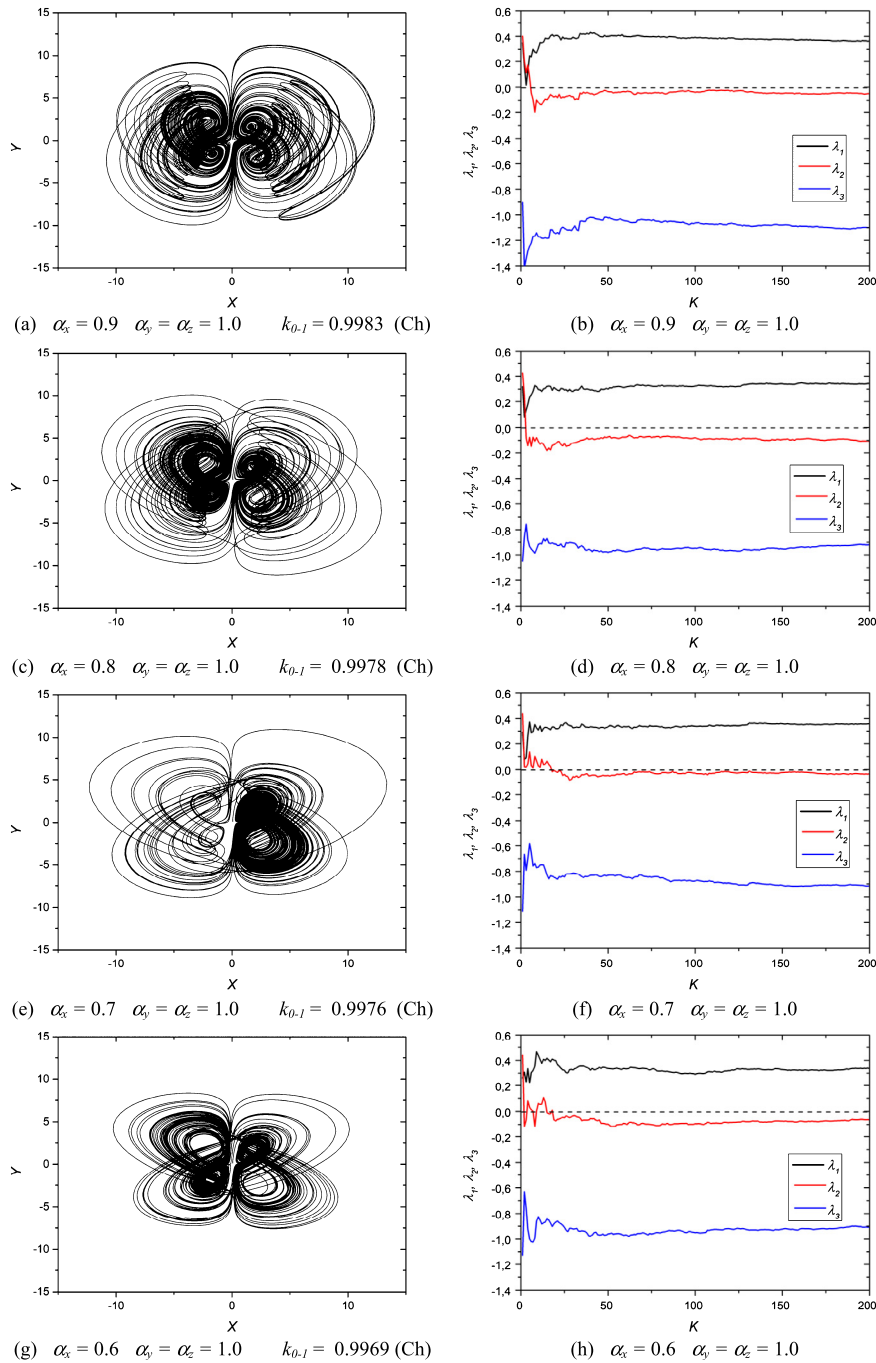


Fig. 6. Four Wing Chaotic System LES results. (a) Incommensurate order $\alpha_x = 0.9\alpha_y = \alpha_z = 1.0$ phase plot and (b) LES; (c) Incommensurate order $\alpha_x = 0.8\alpha_y = \alpha_z = 1.0$ phase plot and (d) LES; (e) Incommensurate order $\alpha_x = 0.7\alpha_y = \alpha_z = 1.0$ phase plot and (f) LES; (g) Incommensurate order $\alpha_x = 0.6\alpha_y = \alpha_z = 1.0$ phase plot and (h) LES.

systems and systems without Jacobian determination, not found before. The evolution of these systems is evaluated with predictor-corrector Adams-Bashforth-Moulton algorithm. Three systems are studied to explore this method for LCE evaluation: a Jerk System, a Financial System and a Four Wing Dynamic System. Commensurate and incommensurate fractional order is explored. The results showed good agreement with respect to phase diagram observation and literature. Jerk system LCE are evaluated with classical approach, and 0-1 Test for Chaos is also employed for all systems, commensurate and incommensurate conditions, showing good agreement with Cloned Dynamics method. Great dependence on simulation and LCE evaluation time parameter T_{Clone} and K are observed and care must be taken when choosing this time parameter,

Table 4Jerk System - LCE asymptotic values and k_{0-1} for 0-1 test.

	α_1	α_2	α_3	λ_1	λ_2	λ_3	k_{0-1}
Comm.	1.0	1.0	1.0	0.1899	0.0413	-0.4246	0.9866
Comm.	0.9	0.9	0.9	-0.0042	-0.0267	-0.4656	0.0758
Comm.	0.8	0.8	0.8	-0.0047	-0.2864	-0.3634	0.0151
Comm.	0.7	0.7	0.7	-0.0527	-0.0528	-0.2763	-0.1425
Incomm.	0.9	1.0	1.0	0.1744	0.0246	-0.4600	0.9928
Incomm.	0.8	1.0	1.0	0.0810	-0.0162	-0.4017	0.3786
Incomm.	0.7	1.0	1.0	-0.0395	-0.0888	-0.2442	0.1543
Incomm.	0.6	1.0	1.0	0.0179	-0.0971	-0.2816	0.1329

Table 5Financial System - LCE asymptotic values and k_{0-1} for 0-1 test.

	α_1	α_2	α_3	λ_1	λ_2	λ_3	k_{0-1}
Comm.	1.0	1.0	1.0	0.0891	0.0058	-0.4348	0.9984
Comm.	0.9	0.9	0.9	0.1133	-0.0030	-0.3370	0.9974
Comm.	0.8	0.8	0.8	-0.0407	-0.0406	-0.2648	0.1325
Comm.	0.7	0.7	0.7	-0.2125	-0.4539	-0.4539	0.2491
Incomm.	0.9	1.0	1.0	0.1445	0.0012	-0.4419	0.9980
Incomm.	0.8	1.0	1.0	0.1235	-0.0026	-0.3412	0.9975
Incomm.	0.7	1.0	1.0	0.0935	-0.0476	-0.2900	0.9986
Incomm.	0.6	1.0	1.0	-0.0678	-0.0680	-0.3308	0.1037

Table 6Four Wing System - LCE asymptotic values and k_{0-1} for 0-1 test.

	α_1	α_2	α_3	λ_1	λ_2	λ_3	k_{0-1}
Comm.	1.0	1.0	1.0	0.3358	0.0133	-1.2608	0.9980
Comm.	0.9	0.9	0.9	0.2684	-0.0259	-0.9699	0.9982
Comm.	0.8	0.8	0.8	-0.1543	-0.1531	-0.801	0.1352
Comm.	0.7	0.7	0.7	-0.4825	-0.4818	-0.6942	0.1752
Incomm.	0.9	1.0	1.0	0.3623	-0.0501	-1.0984	0.9983
Incomm.	0.8	1.0	1.0	0.3460	-0.1048	-0.9196	0.9978
Incomm.	0.7	1.0	1.0	0.3598	-0.0345	-0.916	0.9976
Incomm.	0.6	1.0	1.0	0.3371	-0.0628	-0.9076	0.9969

and LCE results must be compatible with attractor nature (fixed point, limit cycle or strange attractor) observed in phase plot.

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